

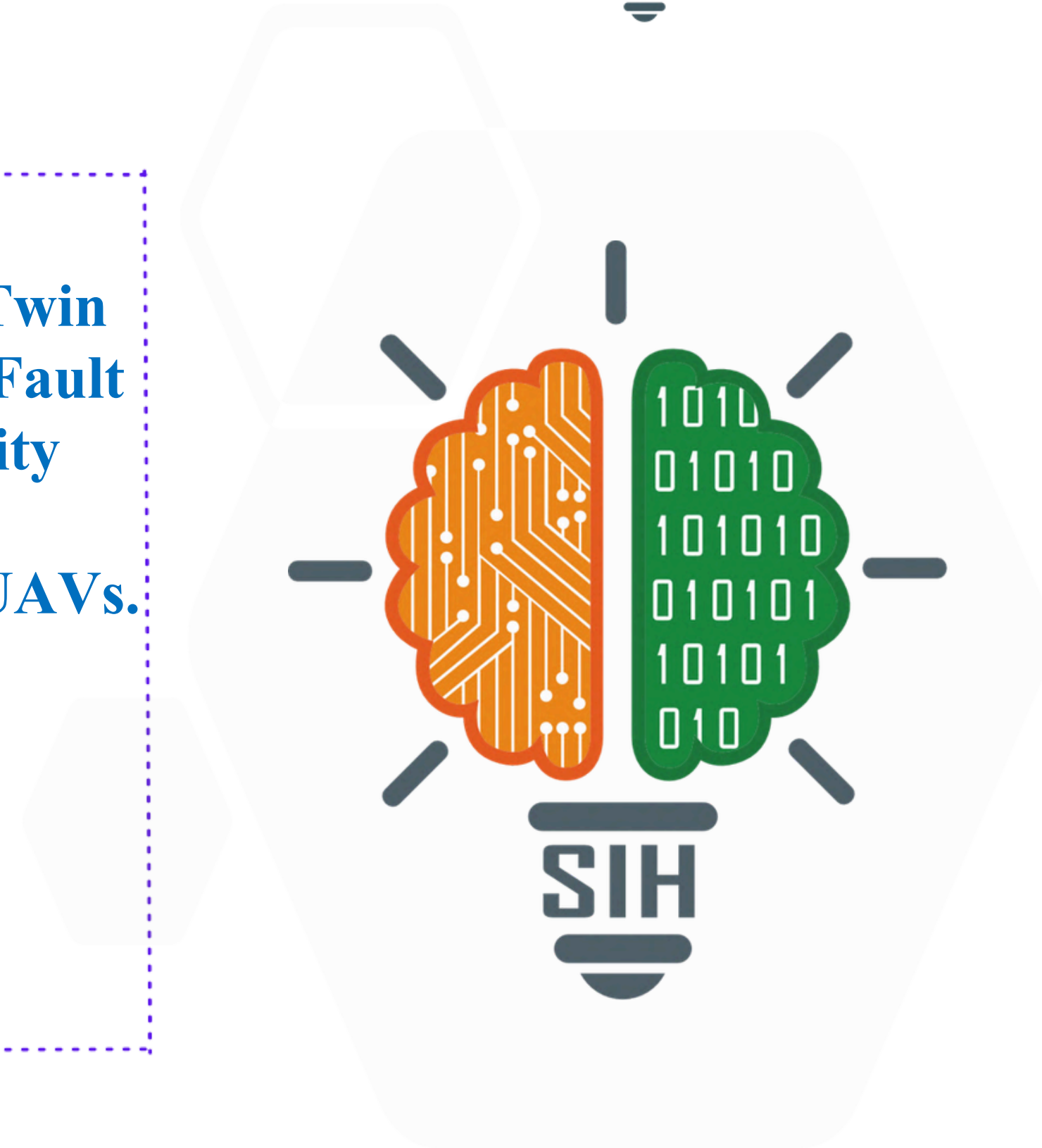
SMART INDIA HACKATHON 2026



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Problem Statement ID : **SIH26054**
Problem Statement Title : **AI-Enabled Real-Time Digital Twin System for Health Monitoring, Fault Prediction and Mission Reliability Enhancement of Aero Piston Engines used in MALE UAVs.**

Theme : **Robotics and Drones**
PS Category : **Software**



PROPOSED SOLUTION

HERMES

A physics-informed digital twin pairing high-frequency telemetry with thermodynamic models to pinpoint anomalies, compute RUL, and recommend safer mission paths.

Addressing the Problem



Beyond Static Thresholds

Mission-aware anomaly detection under changing loads



Physics + AI Fusion

Physics model + live telemetry AI diagnostics



Health-Aware Decisions

Predict degradation and recommend safer missions



Individual Engine Fingerprinting

Creating a **unique** behavioural baseline **each individual engine**

REAL-WORLD CASE — MQ-1B PREDATOR | DJIBOUTI | 2011

Without Hermes - Crash

- **3 Prior Flights:** Chronic low-oil warnings dismissed as transient sensor noise.
- **Mishap Sortie:** Erratic oil pressure followed by sharp **#3 Cylinder EGT drop**.
- **Outcome:** Catastrophic in-flight engine seizure & total UAV loss.

Our Intervention - Saved

- **Early Anomaly Detection:** Flags thermodynamic residual mismatch **45 mins** prior to critical drop.
- **XGBoost** flags oil starvation fault; **LSTM** forecasts critical failure within 30 mins.
- **Outcome:** Recommends immediate RTB & throttled envelope -> **Asset Recovered**.

Innovation and Uniqueness

"What-If" Counterfactual Mission Sandbox

Simulating **"what if"** scenarios, with different flight profiles

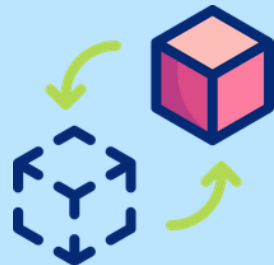
OUTCOME

HEALTH INFORMED
MISSION DECISION

SAFER OPTIMIZED
OPERATIONS

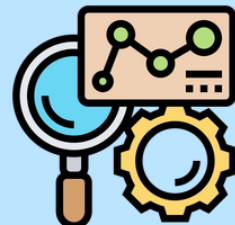
DIGITAL TWIN CORE

- PINN + ODE, MVEM
- Unscented Kalman Filter (UKF)
- Real time residual tracking



AI DIAGNOSTICS + PROGNOSTICS

- XGBoost fault classifier
- Isolation Forest anomaly check
- Monte Carlo Dropout LSTM



MISSION RISK EVALUATION

- 5-point safety threshold check
- 5-min fast forward simulation
- LSTM rescores mission RUL



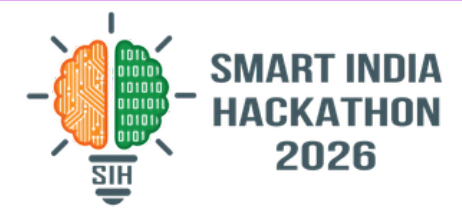
Unknown Fault Discovery

Confidently **state** an **unknown error**, rather than **misclasify** as something else

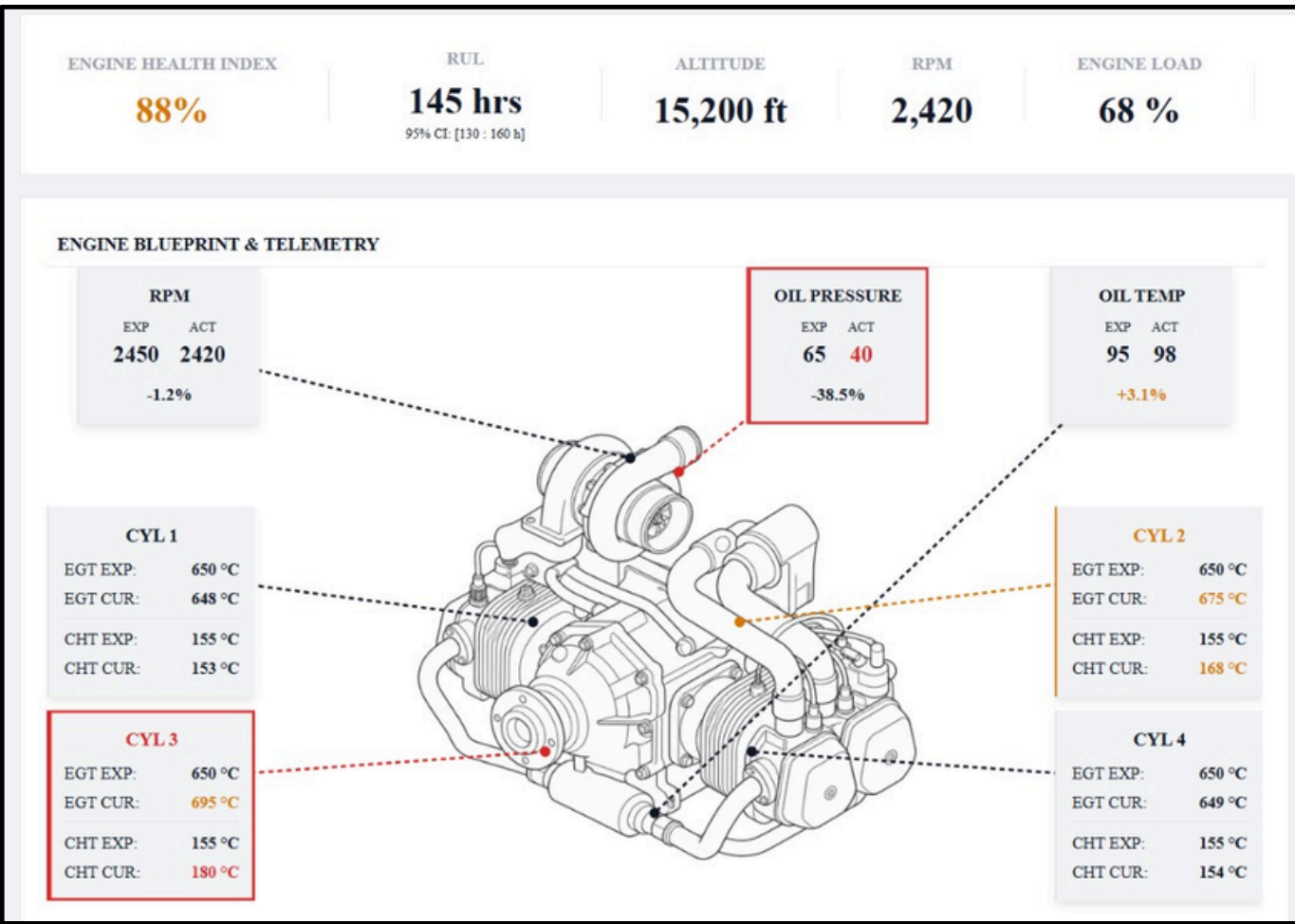
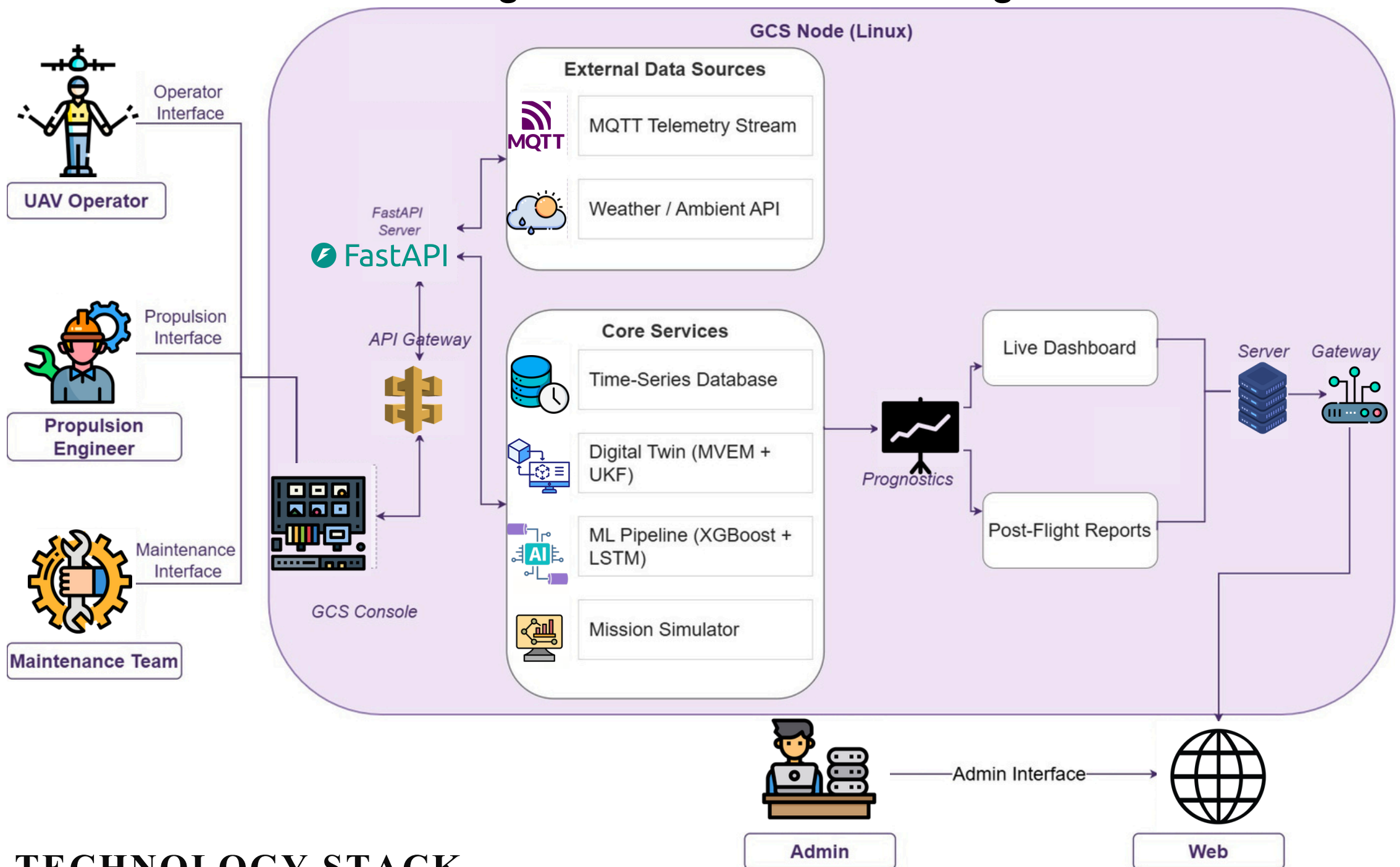


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TECHNICAL APPROACH



UAV Digital Twin & Health Monitoring Architecture



PROTOTYPE DEMO



[Click here to watch demo video](#)

TECHNOLOGY STACK

Synthetic Data + Physics



Backend API



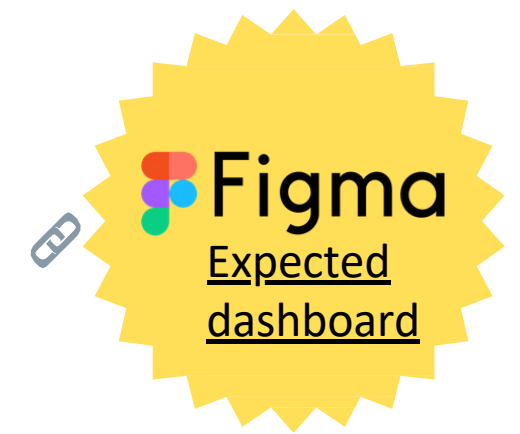
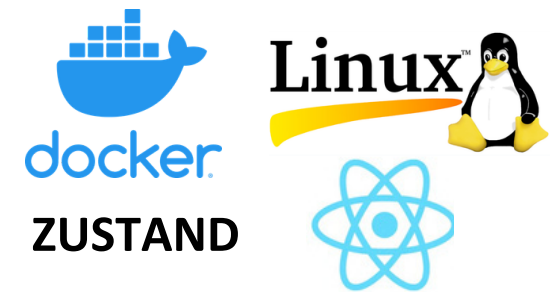
Machine Learning and AI



Databases



Dashboard & Dployment



FEASIBILITY AND VIABILITY

Data Feasibility

- Overcomes the **absence of open-source aero-piston failure data**
- Physics-calibrated **MVEM** and **ISA** injects controlled degradation, lubrication loss, CHT/EGT runaway, and sensor drift; **across high-altitude & hot-day profiles**

Legal Feasibility

- Operates as a passive **GCS-integrated Class-C** advisory
- No **FADEC/ECU** control feedback
- Maintaining **STANAG 4671** safety boundaries and air-gapped data isolation.

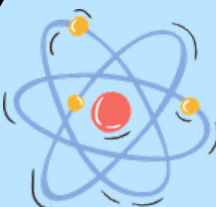
Software Feasibility

- Modular microservices for low-latency telemetry ingestion, time-series storage, high-frequency analytics
- Containerized **deployment on Linux-based military GCS nodes.**

RISK

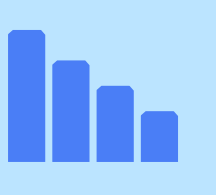
MITIGATION

Real UAV/engine fault data is scarce, **expensive** and often proprietary



- Physics-based** fault generator injects parameter-level faults
- Producing **labeled trajectories** directly from the physics model.

A single large sensor deviation can be a **false alarm**, not a real fault



- Persistence + trend scoring**
- Combines multi-sensor persistence, trend, and normalized residual magnitude
- Flag degradation** reliably.

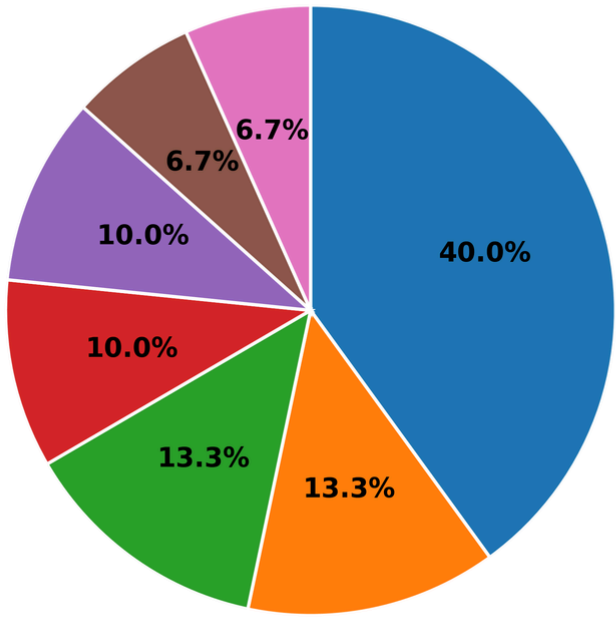
Black-box ML lacks verifiable explanations, hindering safety audits and **engineer trust.**



- Physics-informed anomaly detection(PINN)** constrains predictions by physical laws
- Provides root-cause attribution instead of vague statistical outputs.

HARDWARE VIABILITY

- Reuses existing telemetry**, minimizing **SWaP** and integration costs across UAV platforms.
- Adds interface hardware **only when required** by UAV architecture.
- Minimizes** payload, power, wiring, and **integration complexity** during deployment.



R&D Budget 1.5Cr

Personell & AI Engineering	40%
Hardware & HIL Rig	13.3%
Testing & Qualification	13.3%
Software Compute	10%
Field Pilot & Deployment	10%
Security & Documentation	6.7%
Contingency Reserve	6.7%



Feasible to Prototype

Demonstrable on realistic synthetic telemetry calibrated against reference turbo aero-piston specs (Rotax 914 F / Rotax 915 iS envelope).



Physically Grounded

Coupled **Thermodynamic Model + PINN** analytics ensures outputs respect **real combustion**, thermal balance, and lubrication kinetics.



Safety by Design

Pure advisory decision-support architecture; maintains **GCS air-gap security** with **zero unauthorized write-back access to flight controls**

IMPACTS AND BENEFITS

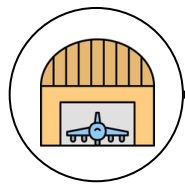


Operational Impact on MALE-UAV Missions



MISSION SUCCESS RATE

Predictive, mission-aware risk scoring
raising odds of a planned mission
completing **without any engine failures**



GROUNDING REDUCTIONS

Early detection of degradation
(cooling, lubrication, combustion)
catching faults before
unscheduled groundings



ROOT CAUSE CLARITY

Physics-oriented diagnostic advice,
showing pilots & technicians the **exact**
sub-system fault(eg., injector imbalance
vs sensor failure)

KEY TAKEAWAYS

- Zero unscheduled groundings
- Prevention of in-flight engine loss
- RUL Driven maintenance
- GCS Ready integration

OPERATIONAL IMPACTS



Preemptive Fault Detection

- Identifies thermal divergence & mechanical wear hours before static thresholds are breached;
- Across **high-altitude and hot-weather envelopes**



Elimination of False Alarms

- Tail-specific engine baselines adapt to each **engine's individual wear profile**
- Preventing false aborts triggered by generic avg. limits



Sensor Drift vs. Engine Fault Isolation

- Separates thermocouple/pressure transducer drift from genuine mechanical seizure
- Preventing unnecessary mission aborts

BROADER ASPECTS



Condition-Based Maintenance Overhaul

- Eliminates arbitrary calendar-hour teardowns
- Scheduling maintenance based on verified **Remaining Useful Life (RUL)** and degradation trends.



Fleet-Wide Scalability

- Modular CAN/telemetry architecture integrated into **Ground Control Stations (GCS) & engine test rigs**
- No modifying airframe wiring required



Strategic Defence & ISR Assurance

- Guaranteed propulsion reliability for high endurance missions (maritime surveillance, border patrol, and communication relay)
- Reliability where engine failure means total mission compromise.



Asset Preservation & Safe Recovery

- Eliminates engine freeze and mid-air power loss
- Ensures damaged engines retain enough throttled capability for a safe recovery or Return-to-Base (RTB) instead of a hull loss.



Lifecycle Fleet Economics

- Extends engine operational life cycle,
- Reduces premature component swaps
- Slashes maintenance labor costs across the entire UAV fleet

OUR UNDERSTANDING:

Aspect	Conventional Systems	Research Gaps	Proposed System (Hermes)
Monitoring	Fixed threshold alarms	Ignores altitude, load & phase context	Physics-aware residual analysis (PINN + UKF)
Fault Detection	Single-sensor breach	No sensor vs engine vs model separation	Multi-sensor triage + zero-day anomaly detection (Isolation Forest)
Diagnostics Depth	Thermodynamic alerts only	Faults visible only after degradation starts	Acoustic + vibration fusion detects sub-component faults hours earlier
Health Assessment	Binary pass/fail	Point RUL estimates; no uncertainty	EHl (0–100) + RUL with 95% CI (MC Dropout LSTM)
Adaptability	Fleet-average baselines	Ignores per-engine manufacturing variance	Per-engine one-class fingerprinting
Mission Intelligence	No engine-aware planning	Health prediction decoupled from mission	Future mission + constraint-aware what-if prescriptive engine
Robustness	Vulnerable to spoofing	Can't distinguish spoofed vs real faults	Physics-grounded detection + thermodynamic impossibility flagging
Training Data	Needs real fault records	Real fault data rare & proprietary	Synthetic fault generator with physics-based degradation injection

References

[1] [DJIBOUTI Accident Case Study](#)

[2] [Peringal, A., Mohiuddin, M. B., Haddad, A. G., & Muthusamy, P. K. \(2025\). Reliable prediction of remaining useful life for aircraft engines: An lstm-based approach with conservative loss function. In AIAA SCITECH 2025 Forum \(p. 1909\).](#)

[3] [Nath, K., Kumar, V., Smith, D. J., & Karniadakis, G. E. \(2026\). A digital twin for diesel engines: Operator-infused physics-informed neural networks with transfer learning for engine health monitoring. Engineering Applications of Artificial Intelligence, 170, 114052.](#)

Datasets for references:

[1] [NGAFID_MC_20210917](#)

[2] [A Large-Scale Annotated Multivariate Time Series Aviation Maintenance Dataset from the NGAFID](#)